Software Engineering

Requirements Engineering

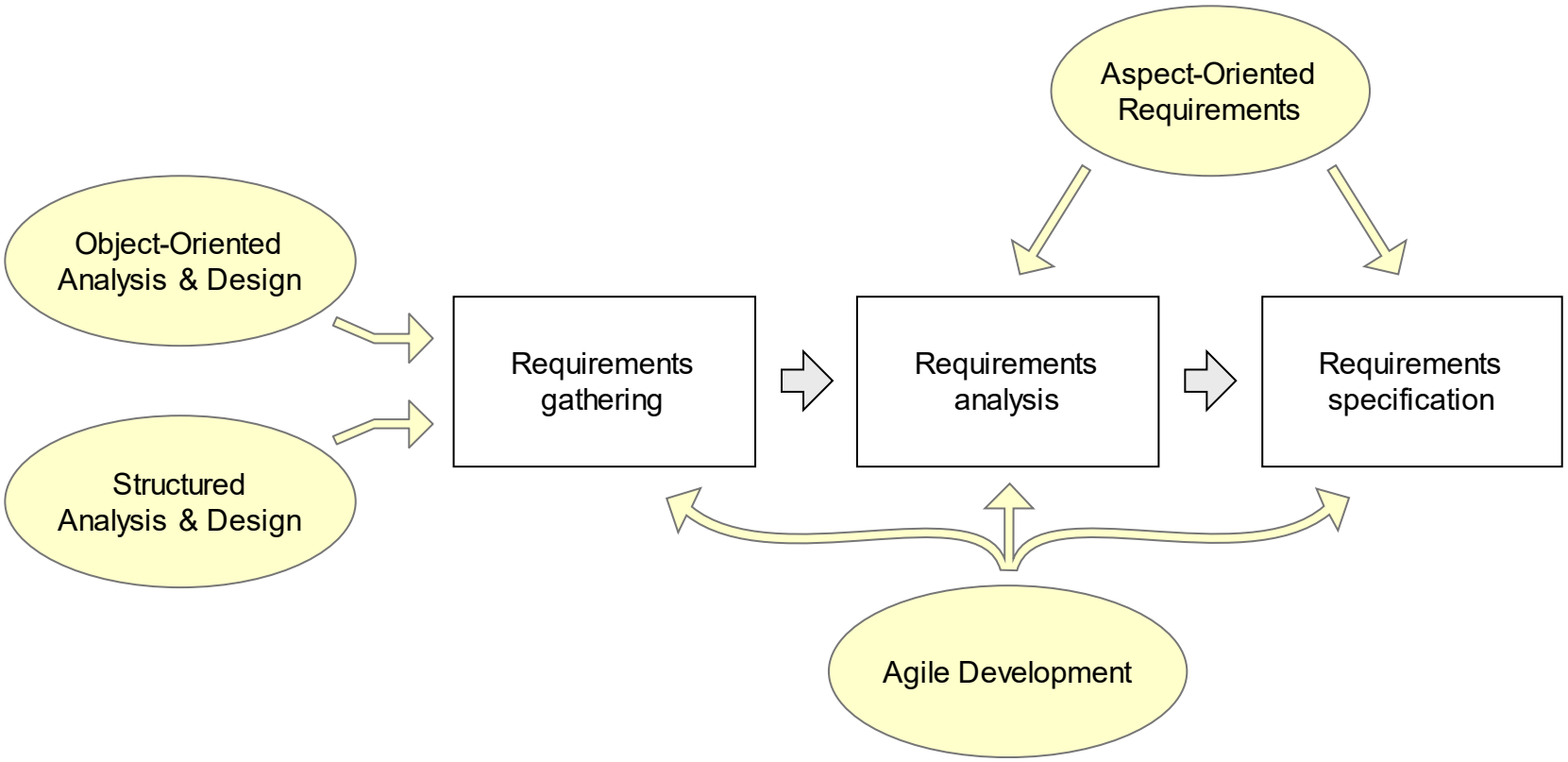
Topics

- Requirements Engineering Components
- Requirements and User Stories
- Agile Methods for Effort Estimation and Work Organization
- Requirements Analysis
- Acceptance Tests
- Types of Requirements

Software Requirements

- A **requirement** specifies the business functions that the user will be able to perform using the system-to-be in different “situations” or “contexts”, and the kind of experience the user will have during this work
  - Other concerns, such as how the system will manage the resources (computing, network, …), how the system will manage and protect user’s data, etc.
- User requirements will often be high-level, vague and incomplete. They are more like high-level goals, or business goals, rather than software requirements needed by the developer
- When trying to achieve a given high-level goal, we will need to consider what matters, what are the important parameters, so that we can derive the detailed technical requirements
- Only based on deeper understanding of detailed issues, we can identify important "scenarios" or "situations" and identify what parameters should be considered in each situation
- Then using these parameters, we decide what the system should do, or how to respond to this situation (i.e., inputs)

Requirements Process



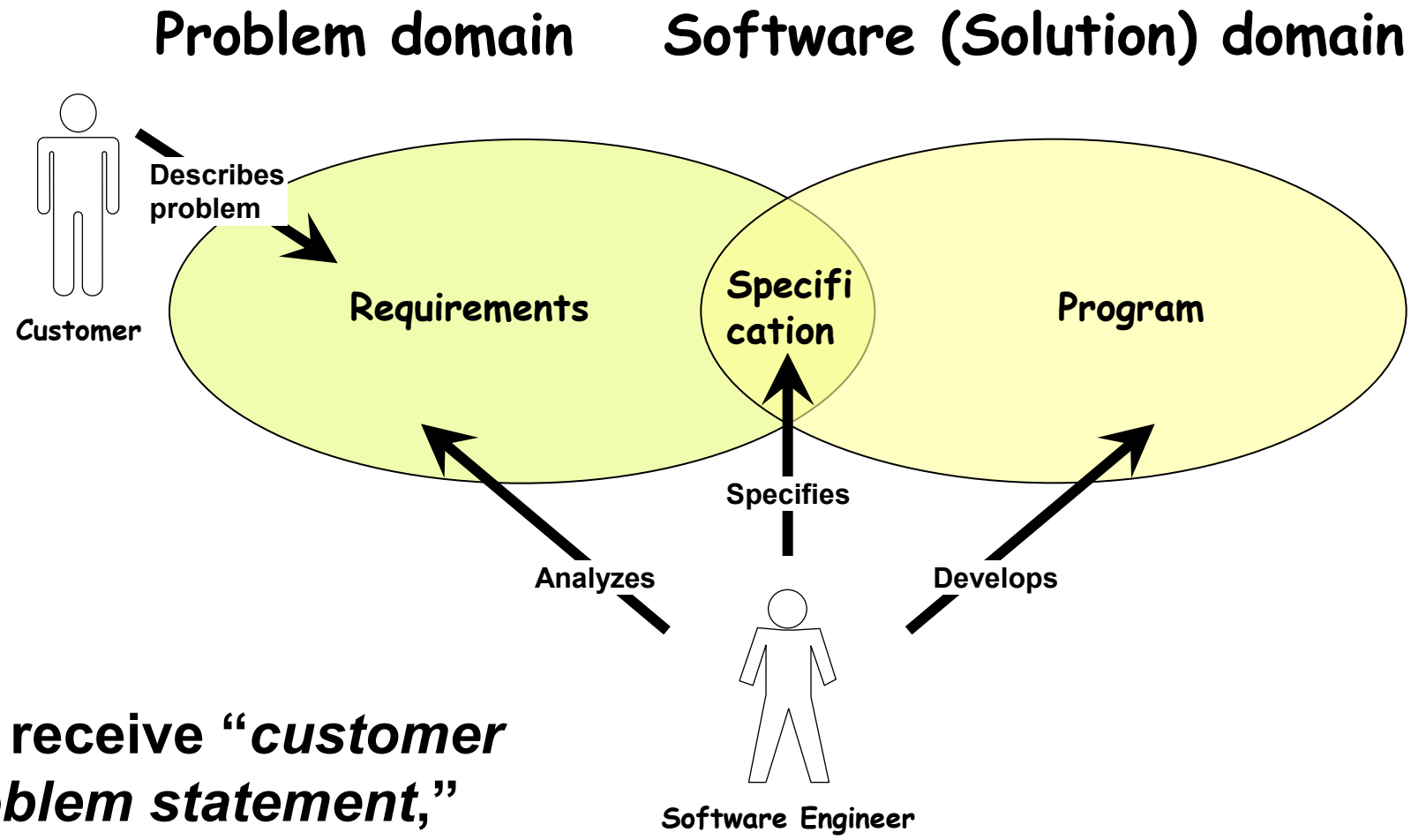
Requirements Engineering Components

- Requirements gathering
  - (a.k.a. “requirements elicitation”) helps the customer to define what is required: what is to be accomplished, how the system will fit into the needs of the business, and how the system will be used on a day-to-day basis
- Requirements analysis
  - refining and modifying the gathered requirements
- Requirements specification
  - documenting the system requirements in a semiformal or formal manner to ensure clarity, consistency, and completeness

Practical Requirements Engineering

- Test your idea in practice and use the result in further work, iterating through these creative and evaluative steps until a solution is reached
  - No one can know all the constraints for a solution before they go through the solving experience
- Define the criteria for measuring the success ("acceptance tests")
- Avoid *random* trial-and-error by relying on domain knowledge (from publications or customer expertise)

Requirements and Specification



We receive "*customer problem statement*," not the requirements!

Requirements Derivation

Detecting that a problem exists is different from *defining* the problem and its causes, and solution constraints.
Depending on the cause, the solution will be different.

Requirements are determined by:

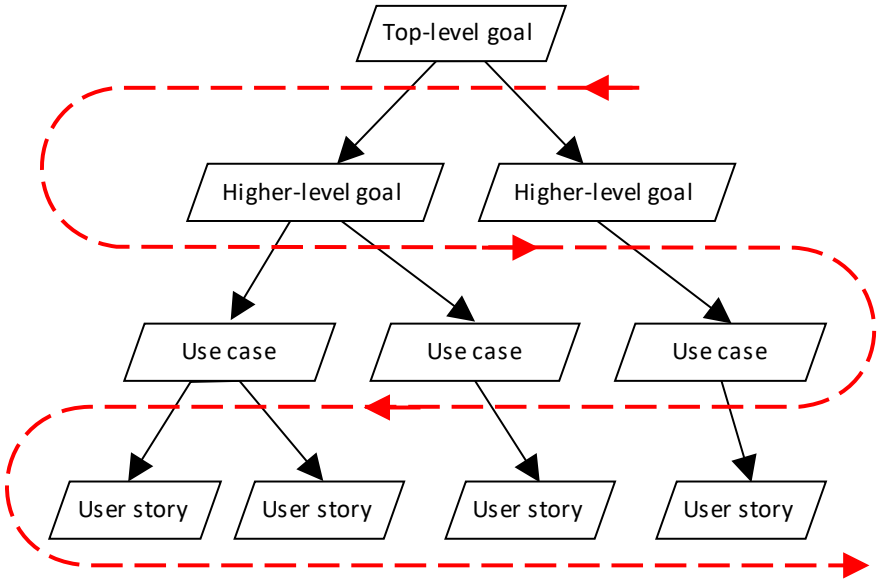
- Judgment about customer's business goals and obstacles that currently are hindering their achievement
- Conditions on solutions imposed by real-world constraints:
  - Physical
  - Social/Cultural
  - Legal
  - Financial
  - …
- Threats created by adversaries

➔ Requirements are **not** simply desires!

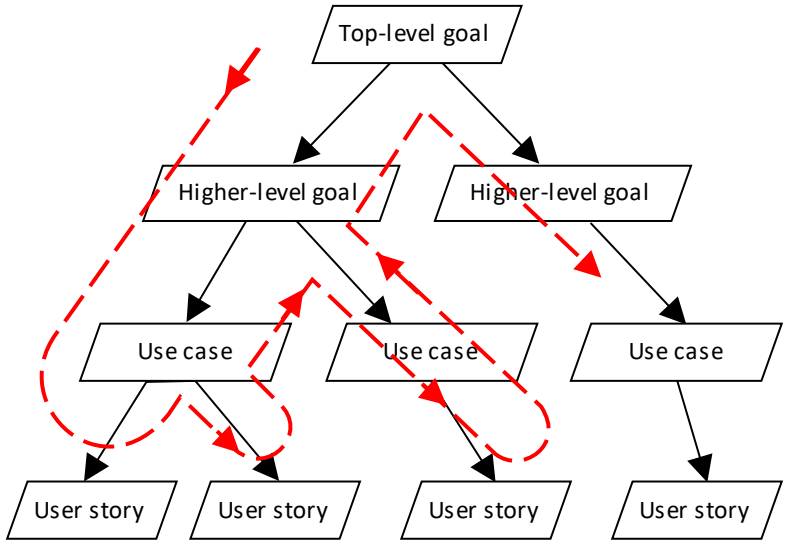
- Requirements are desires *adjusted to real-world constraints and threats*

Decomposition of Business Goals

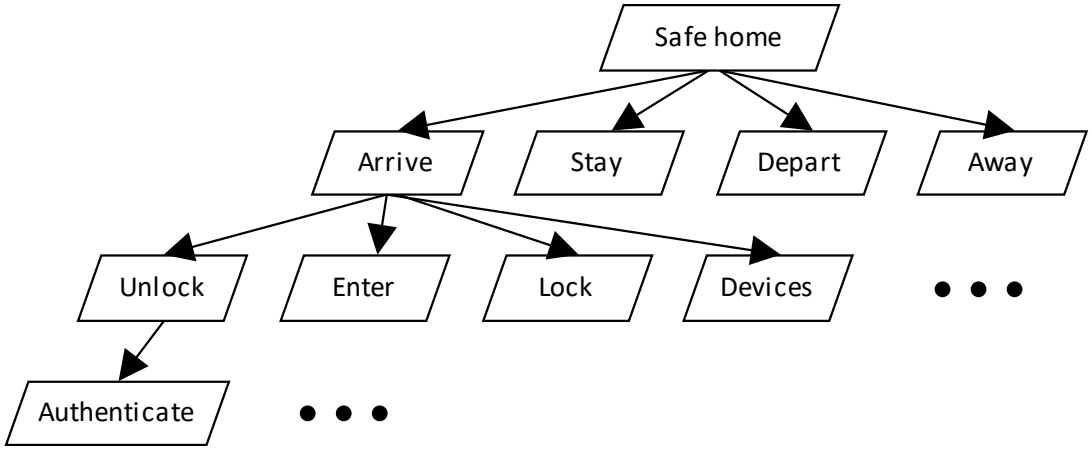
Breadth-first refinement

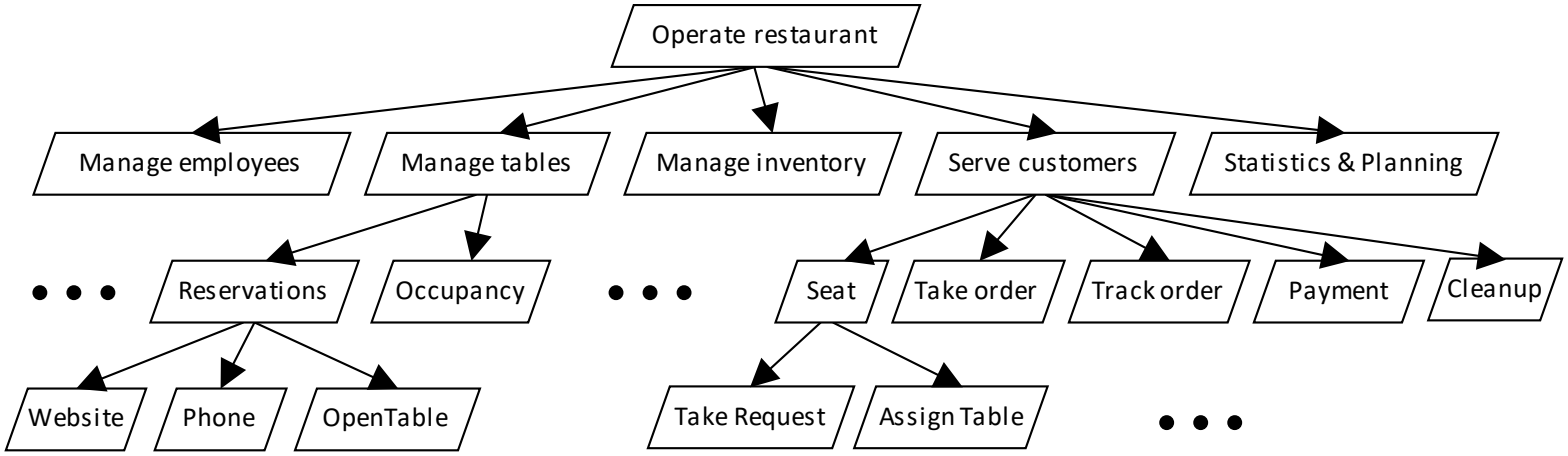


Depth-first refinement



Examples: Safe Home Access Restaurant Automation





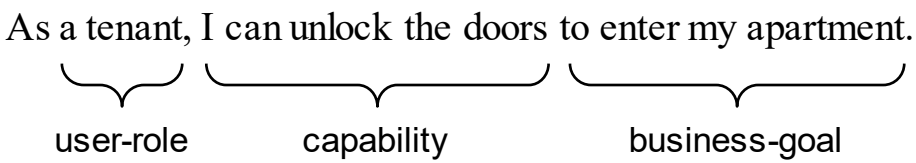
Problem Analysis Examples

- Traffic Monitoring:
  - Problem: User is delayed to work and needs help
  - Cause A: Traffic is unpredictable (predictably high if repeated delays?!)
  - Cause B: User is unfamiliar with the route
  - Cause C: User has a habit of starting late
- Restaurant Automation:
  - Problem: Restaurant is operating at a loss and customers are unhappy
  - Cause A: Staff is encumbered with poor communication and clerical duties, resulting in inefficiencies
  - Cause B: The menu does not match the likeliest customer's taste
  - Cause C: Staff is misbehaving with customers or mismanaging the food supplies
- Health Monitoring:
  - Problem: User is leading unhealthy lifestyle and experiencing health problems
  - Cause A: User is trying to be active but unsure about sufficient level of activity
  - Cause B: User is trying to be active but too busy with other duties
  - Cause C: User cannot find affordable solutions for active lifestyle
  - Cause D: User has poor sleep
  - Cause E: User is aware of issues but not motivated to act

Problem Example: Safe Home Access

- Problem detected:
  inconvenient physical keys or unwanted intrusion
  (plus: operating household devices and minimizing living expenses)
- Analysis of the Causes:
  - User forgets to lock the door or turn off the devices
  - User loses the physical key
  - Inability to track the history of accesses
  - Inability to remotely operate the lock and devices
  - Intruder gains access
- System Requirements: based on the selected causes

Requirements as User Stories

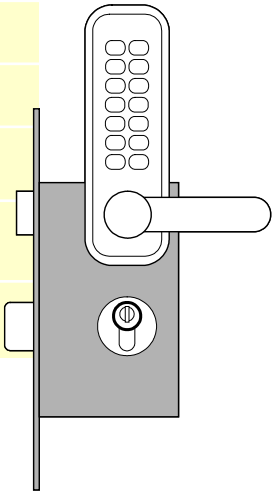


- Preferred tool in **agile methods**.
- Stated in terms of user's goals and capabilities instead of system features
- Written by the customer or user, not by the developer.
- The development effort to implement a story is estimated immediately
- Acceptance tests are written when the story is identified
- Requirements identified only for the next iteration, not for the whole project

Example User Story Requirements

Safe Home Access

Identifier	User Story	Size
REQ-1	As a user, I can be sure that the doors by default will be locked.	**4 points**
REQ-2	As a user, I will be able to unlock the doors using a valid key.	**7 points**
REQ-3	An intruder will not be able to unlock the doors by guessing a valid key; the system will block when it detects a “dictionary attack.”	**7 points**
REQ-4	As a user, I can be sure that the doors will be automatically locked at all times.	**6 pts**
REQ-5	The door keypad will be backlit when dark for visibility.	**3 pts**
REQ-6	Anyone will be able to lock the doors on demand.	**3 pts**
REQ-7	As a user, I will be able to manage additional user accounts.	**10 pts**
REQ-8	As a user, I will be able to view the history of accesses to my home.	**6 pts**
REQ-9	As a user, I will be able to configure the preferences for how my household devices will be activated on my arrival.	**6 pts**



- Note no priorities for user stories
  - Story priority is given by its order of appearance on the to-do list
- Estimated size points (last column) will be described later (also see Chapter 1)
- Compare to IEEE-830 style requirements
  - https://en.wikipedia.org/wiki/Software_requirements_specification

Example User Story Requirements

Restaurant Automation

Identifier	User Story	Size
REQ-1	As a host, I can take a seating request including customer party information, place into the seating queue, and have a table assigned or waiting time estimated.	10 points
REQ-2	As a waiter, I can input customer's order.	5 points
REQ-3	As a waiter, I can add special instructions to an order at the customer's request.	2 points
REQ-4	As a waiter, I can notify the chef of the order without walking to the kitchen.	2 pts
REQ-5	As a waiter, I can view customer's bill and enter their payment information.	7 pts
REQ-6	As a waiter, I will be notified when an order has been completed.	2 pts
REQ-7	As a chef, I can see the queue of orders waiting to be prepared.	3 pts
REQ-8	As a chef, I can mark orders as "In Preparation" and "Complete".	2 pts
REQ-9	As a chef, I can modify the menu to make certain dishes available or unavailable if supplies are limited.	6 pts
REQ-10	As a chef, I can adjust and update the supply inventory to let the manager know of any lacking ingredients.	7 pts

- After requirements analysis:

Identifier	User Story	Size
REQ-2a	As a waiter, I can input orders at different times for customers at the same table.	7 points
REQ-2b	As a waiter, I can input different courses at different times for the same customer.	5 pts
REQ-2c	As a waiter, I can input side plates after the main course is served.	2 pts

Requirements Analysis

- Requirement REQ-3 states that intruders will not be able to succeed with a "dictionary attack," but many details need to be considered and many parameters determined ("business policies")
  - What distinguishes user's mistakes from "dictionary attacks"
    - The number of allowed failed attempts, relative to a predefined threshold value
      - The threshold shall be small, say three ← business policy!
  - How is the mechanical lock related to the "blocked" state?
    - Can the user use the mechanical key when the system is "blocked"?
- Requirement REQ-5 states that the keypad should be backlit when dark
  - Is it cost-effective to detect darkness vs. keep it always lit?
- Etc.

Requirements analysis should not be exhaustive,
but should neither be avoided.

From Requirements to Business Policies

Explicit identification of **business policies** is important for two reasons:

1. Making the need for BP explicit allows involving other stakeholders, particularly the customer, in decision making about the BP solutions to adopt
2. Helps to anticipate potential future changes in the policies, so mechanisms can be implemented in the code that localize the future changes and allow quick substitution of implemented business policies

These issues too important to be left to the programmer to make ad-hoc decisions and hard-code them.

Tools for Requirements Eng.

- Tools, such as user stories and use cases, used for:
  - Determining what exactly the user needs ("requirements analysis")
  - Writing a description of what system will do ("requirements specification")
- Difficult to use the same tool for different tasks (analysis vs. specification)

Analyst's Task: Three Descriptions

The ***requirement***



The ***problem domain***



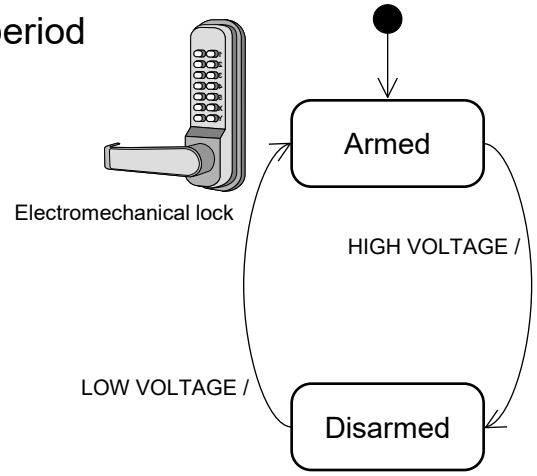
The ***specification***

What user wants:

When valid keycode entered & Unlock pressed, open the lock;

Automatically lock after a period of time.

How problem domain behaves:



What software-to-be will do (at interface):

If entered number matches one of stored numbers **&** Button-1 pressed, put HIGH voltage on Output-port-1;

Start a timer countdown;

When the timer expires, put LOW voltage on Output-port-1.

Concern:

It is not obvious that this is the only or even "correct" solution to the requirement-posed problem.

Problem Frames tell us what each description should contain and how to verify the concern.

Acceptance Tests

- Means of assessing the *project success criteria*
  - The requirements are met as expected
- Conducted by the customer throughout the project, not only at the end
- An acceptance test describes whether the system will pass or fail the test, given specific input values
- We cannot ever guarantee 100% coverage of all usage scenarios, but *systematic approach* can increase the *expected* degree of coverage

A Story … ☺

Dear Mr. Architect,

Please design and build me a house. I am not quite sure of what I need, so you should use your discretion.

My house should have between two and forty-five bedrooms. Just make sure the plans are such that bedrooms can be easily added or deleted.
When you bring the blueprints to me, I will make the final decision of what I want. Also bring me the cost breakdown for each configuration so that I can arbitrarily pick one.

A Story … (Cont'd)

Keep in mind that the house I ultimately chose must cost less than the one I am currently living in. Make sure, however, that you correct all the deficiencies that currently exist in my house (the floor of my kitchen vibrates when I walk across it, and the walls don't have nearly enough insulation in them).

Also keep in mind as you design this house that I wish to keep yearly maintenance cost as low as possible. This should mean the incorporation of extra-cost features like aluminum or vinyl siding. If you chose not to specify aluminum, be prepared to explain in detail.

A Story … (Cont'd)

Please take care that modern design practices and the latest materials are used in construction of the house. The house should be really nice.
However, be alerted that the kitchen should be designed to accommodate among other things, my 1952 Gibson refrigerator.

To assure that you are building the correct house for our family, make sure that you contact each of the children and also the in-laws. My mother-in-law will have very strong feelings about how the house ought to be designed since she visits with us at least once a year. Make sure that you weigh all these options carefully and make the right decision.
I, however, retain the right to override any decision you come up with.

A Story … (Cont'd)

Please don't bother me with small details right now. Your job is to develop the overall plans for this house. Get the big picture. It is not appropriate at this time to be choosing the color of the carpet. However, keep in mind that my wife likes green.

Also do not worry at this time about acquiring resources to build this house. Your first priority is to develop detailed plans and specifications. However, once I accept these plans, I will expect to have the house under roof within 48 hours.

A Story … (Cont'd)

While you are designing this house specifically for me, keep in mind that sooner or later I will have to sell this house. It should have appeal to potential buyers. Please make sure that before you finalize the plans, there is a consensus of the population in my area that they like the features this house has.

You are advised to run up and look at my neighbor's house he had constructed last year. We like it a great deal. It has many features that we would like to have in our new home, particularly the 75-foot swimming pool. With careful engineering I believe that you can design this into our new house without impacting the construction cost.

A Story … (Cont'd)

Please prepare a complete set of blueprints. It is not necessary at this time to do the real design since these blueprints will be used only for construction bids. Please be advised however, that any increase of cost in the future as a result of design changes will result in you getting your hands slapped.

You must be thrilled to be working on such an interesting project such as this. To be able to use new kinds of construction and to be given such freedom in your designs is something that doesn't happen very often. Contact me as rapidly as possible with your design ideas. I am enthusiastic about seeing what you can come up with.

A Story … (Cont'd)

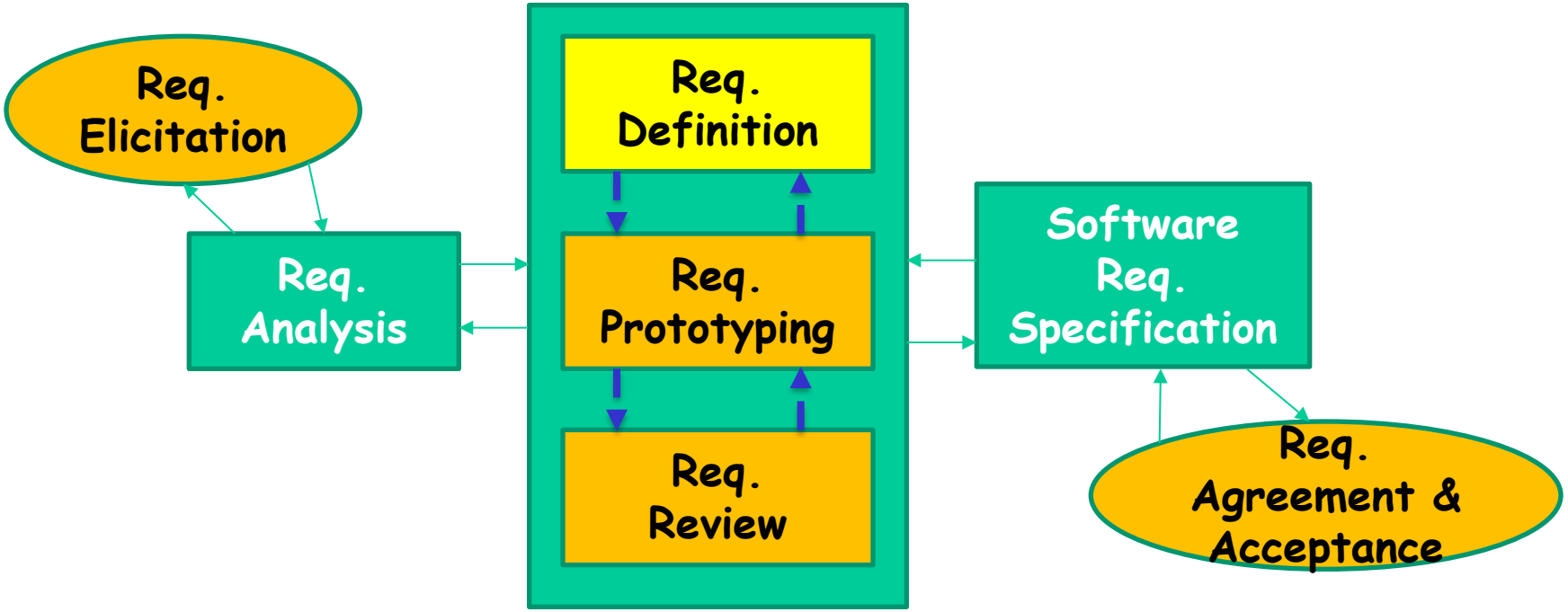
P.S. My wife has just told me that she disagrees with many on the instructions I've given you in this letter. As architect it is your responsibility to resolve these issues. I have tried in the past and have been unable to accomplish this. If you can't handle this, I'll have to look for a new architect.

P.P.S. Perhaps what I need is not a house at all, but a travel trailer. Please advise me as early as possible if that is the case.

Class Exercise

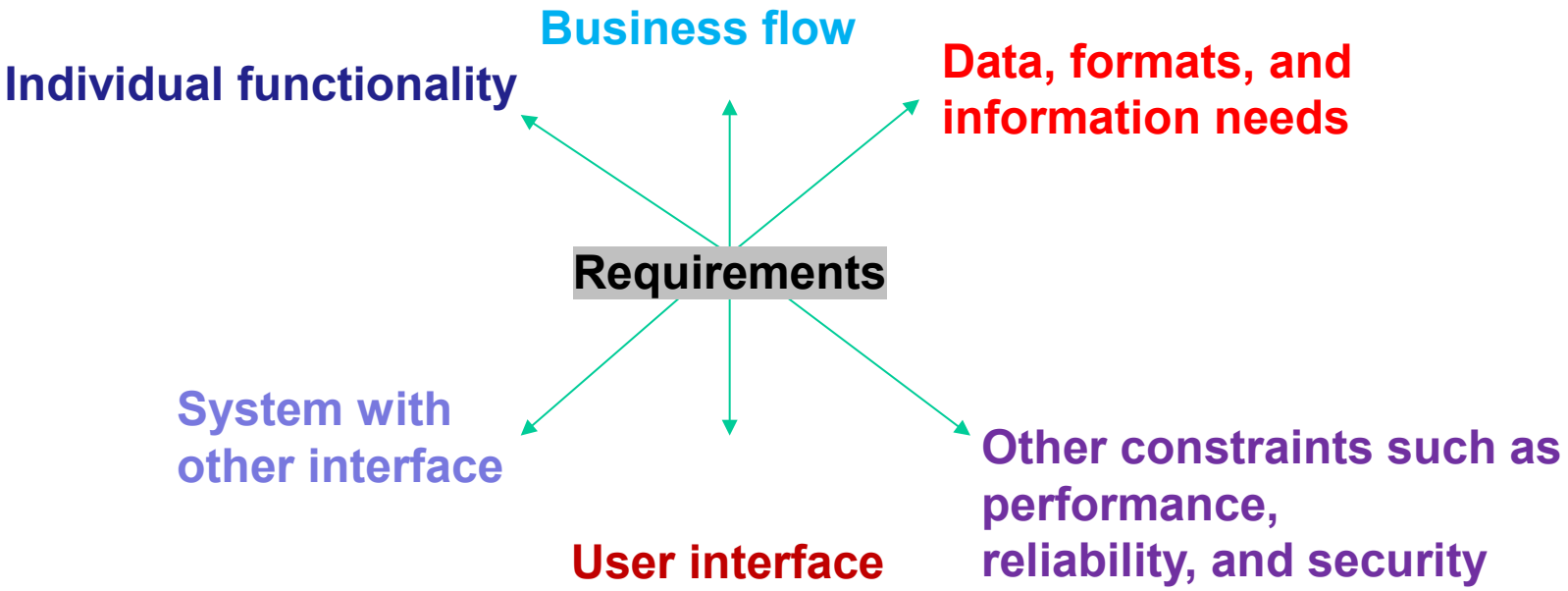
- Write down the requirements/ user stories for the ATM machine.
- Specify the requirements for cash withdrawal, cash deposit, balance check, transfer, etc.

Requirement Engineering Process



Eliciting Detailed Requirements

- More technically savvy requirements analysts should be brought in.



Requirement Analysis

- Even after requirements are elicited and collected, they are still just an unorganized set of data. They must be still analyzed.

- The analysis of requirements consists of two main tasks:
  - Categorizing or clustering requirements
  - Prioritizing the requirements

Categorizing or clustering requirements

Requirements Area	Prefix	Requirement Statement Numbering
Individual functionality	IF	IF-1.1, IF-1.2, If-1.3
Business flow	BF	BF-1, BF-2
Data and data format	DF	DF-1.1, DF-1.2, DF-1.3
User interface	UI	UI-1.1, UI-1.2
Interface to systems	IS	IS-1
Further constraints	FC	FC-1

Requirements Analysis and Clustering with Object-Oriented Use Cases

